



الجامعة الألمانية الأردنية
German Jordanian University

Control Systems 1 Project

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Abstract

This report presents the simulation and analysis of a dynamic control system using MATLAB and Simulink, focusing on optimizing a Proportional-Derivative (PD) controller for improved performance. The goal was to achieve a time constant of 0.5 seconds or less, a damping ratio of 0.707 or greater, and to enhance system stability. Additionally, a Proportional-Integral-Derivative (PID) controller was implemented to ensure zero steady-state error and achieve a 20% improvement in settling time.

Key tasks included modeling the plant transfer function, tuning controller parameters, and conducting root locus and step response analyses. Results demonstrated that the PD controller effectively met transient response objectives, while the PID controller further improved settling time and steady-state accuracy. The study highlights MATLAB's utility in designing and analyzing control systems for academic and practical applications.

Question 1

Theory

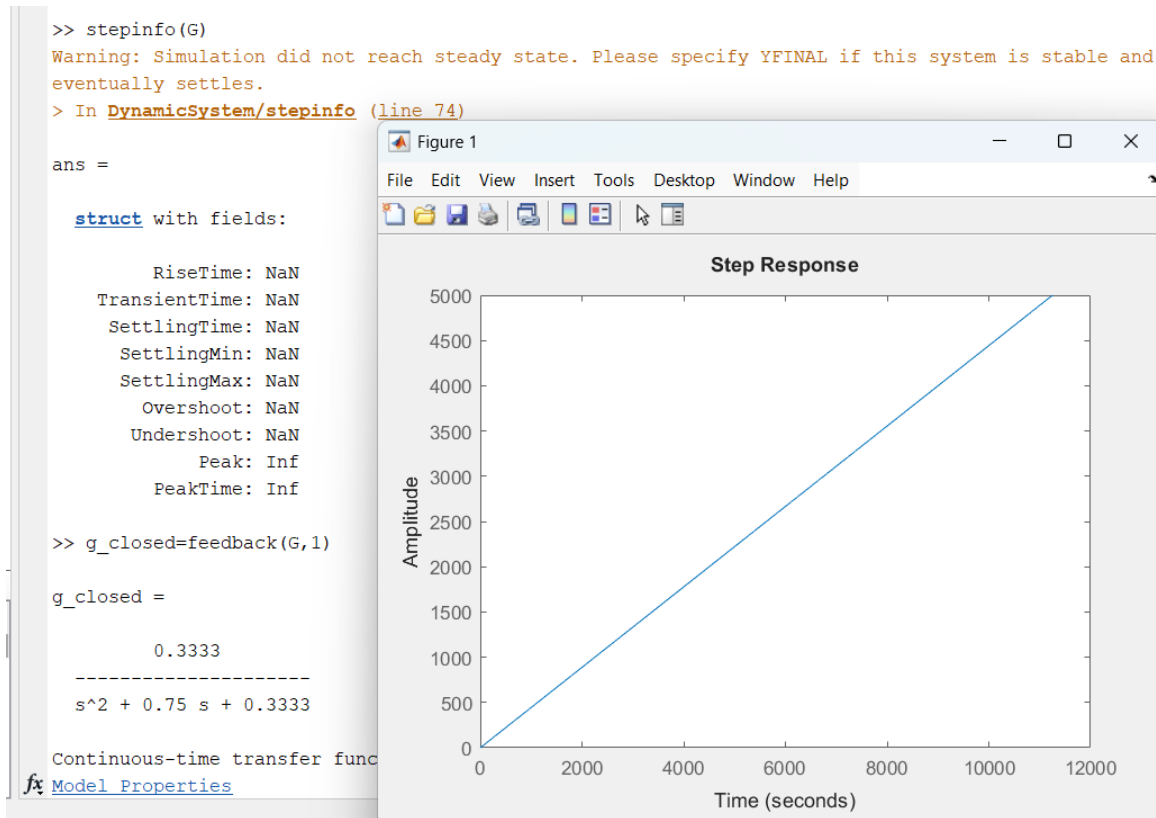
In this question we used the DC motor equation (from chapter 2) to obtain the transfer function.

We took the following steps:

- $K_t / R_a = T_s / E_a = 1$, where $T_{\text{stall}} = 5$ and $E_a = 5$.
- $K_b = E_a / \omega$, where $E_a = 5$, $\omega = 600 / \pi$. Sub, $K_b = 1 / 4$
- $J_m = 16(1/4)^2 + 4(1/2)^2 + 1 = 3$
- $D_m = 32(1/4)^2 = 2$
- Sub. In the DC motor equation. $\theta_m(s) / E_a(s) = (1/3) / (3s^2 + 0.75s)$

Procedures

- i) Inserting the transfer function in MATLAB
- ii) Getting the step for the open loop transfer function:



iii) closing loop transfer function with unity feedback:

```
>> g_closed=feedback(G,1)
```

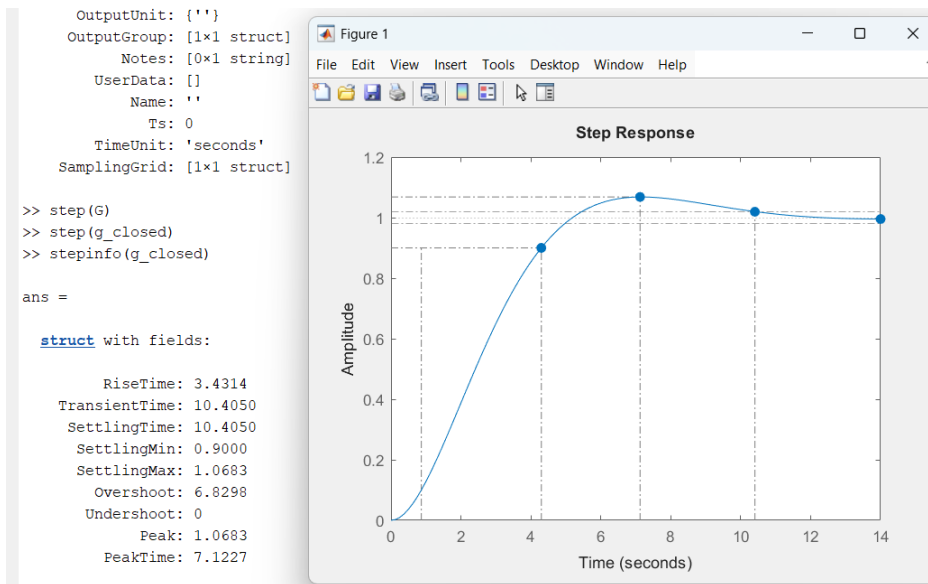
```
g_closed =
```

```

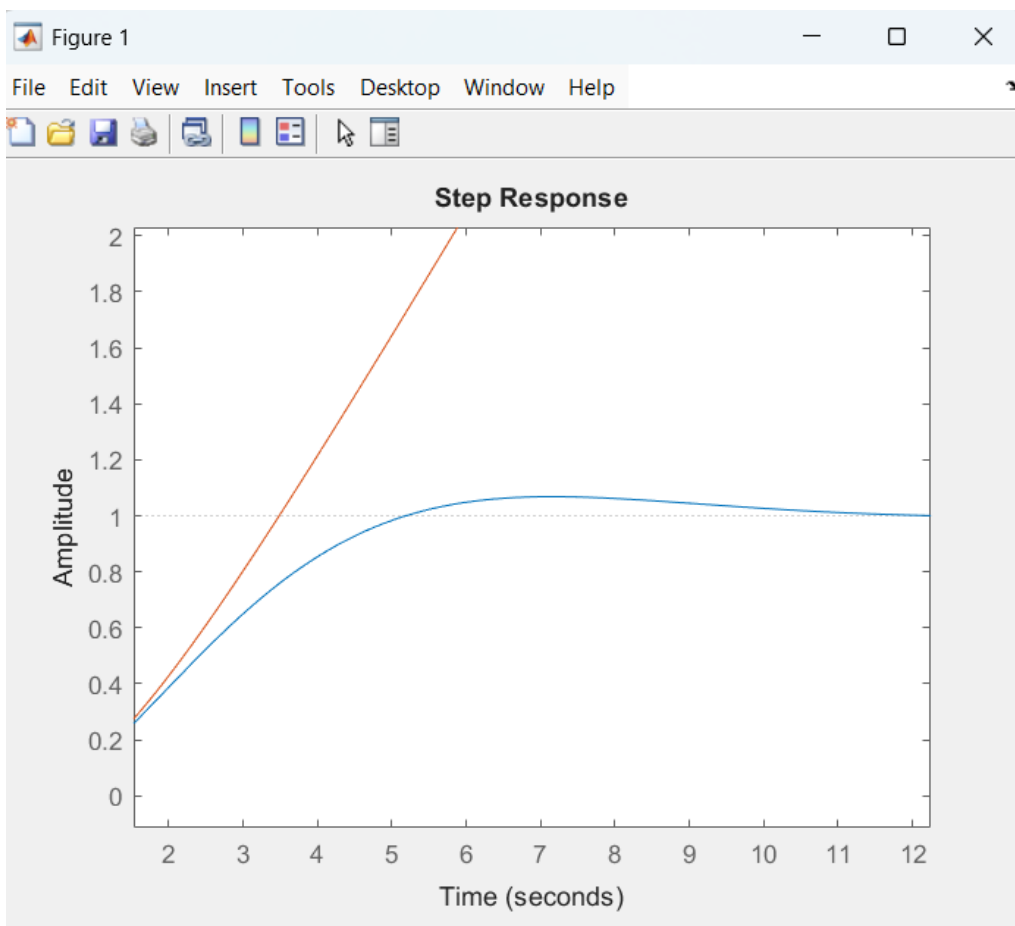
    0.3333
-----
s^2 + 0.75 s + 0.3333

```

Closed loop step:



The following figure compares the step response for the open loop and closed loop:

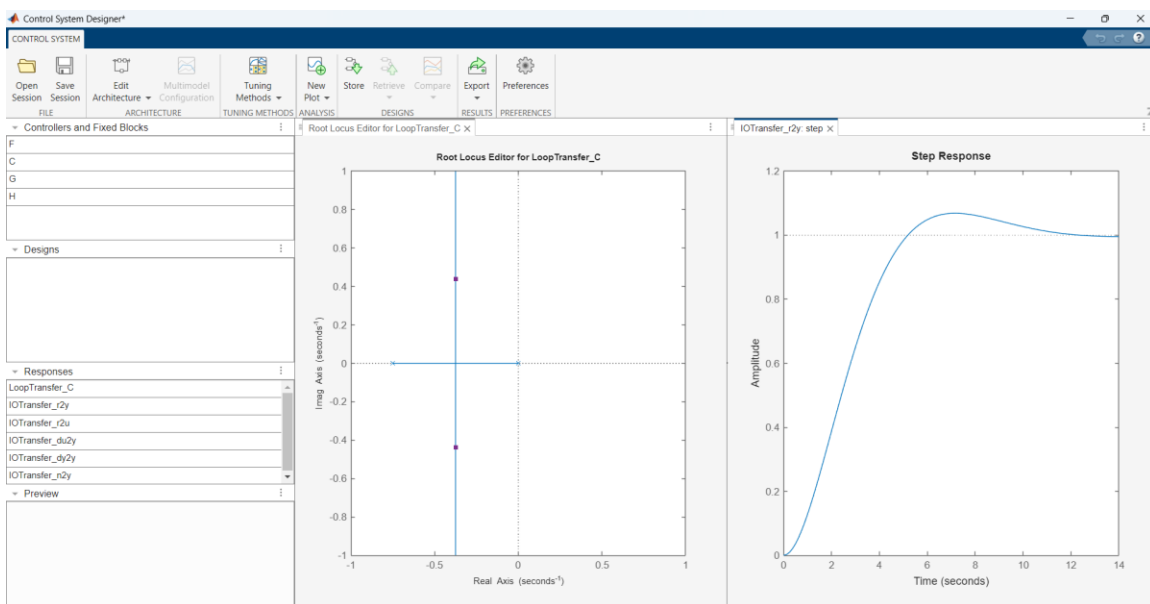


Comparison Table:

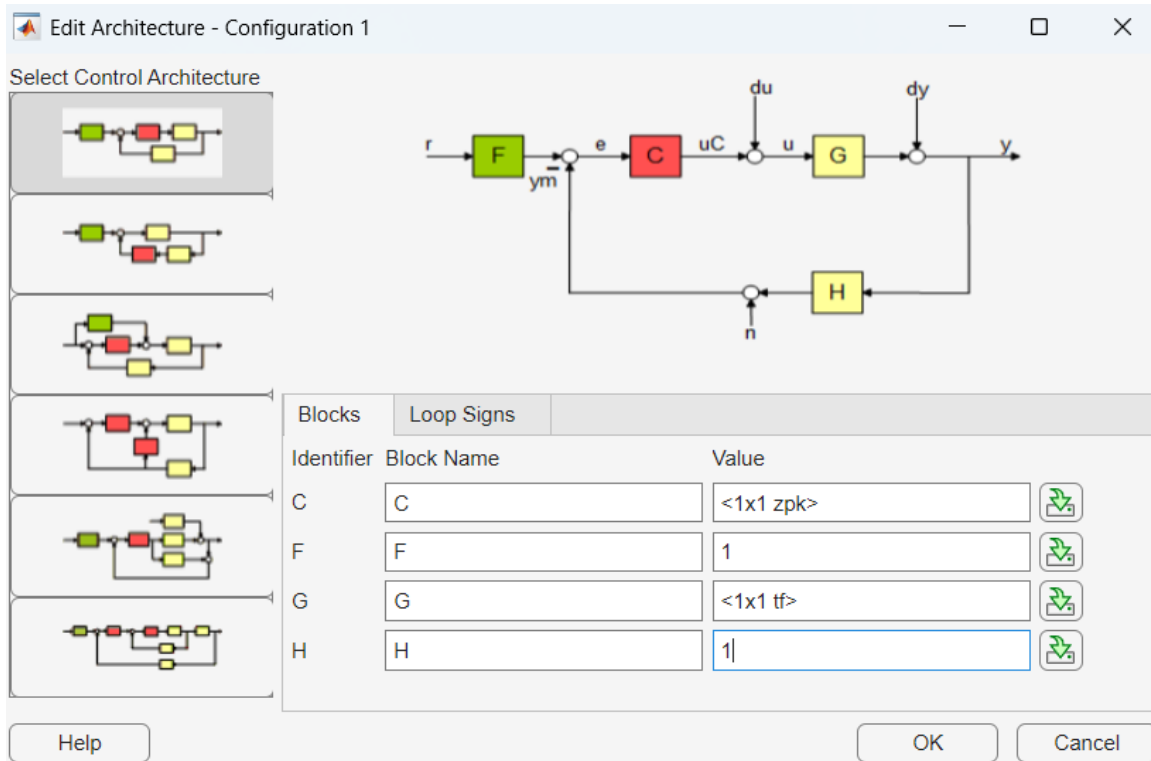
STEP INFO	OPEN LOOP	CLOSED LOOP
RISE TIME	NaN	3.4314
TRANSIENT TIME	NaN	10.4050
SETTLING TIME	NaN	10.4050
SETTLING MIN	NaN	0.9000
SETTLING MAX	NaN	1.0683
OVERSHOOT	NaN	6.8298
UNDERSHOOT	NaN	0
PEAK	Inf	1.0683
PEAK TIME	Inf	7.1227

iv) The current settling time is 10.4050, with 20% improvement we should get settling time to be <8.3240.

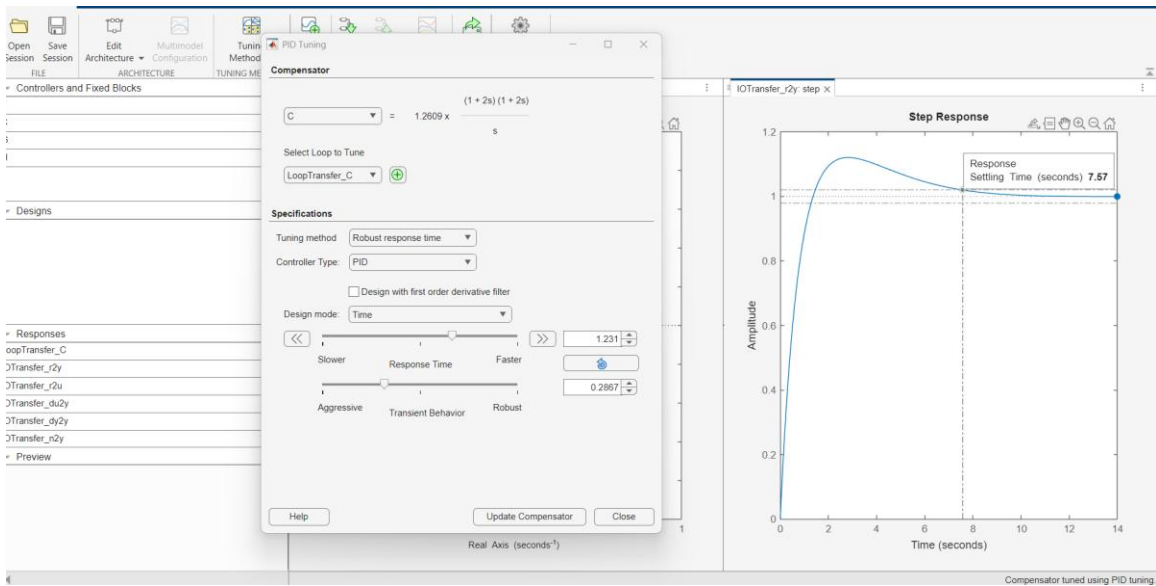
i) We open control system designer:



ii) we set our architecture:



iii) 3- Using PID tuner:

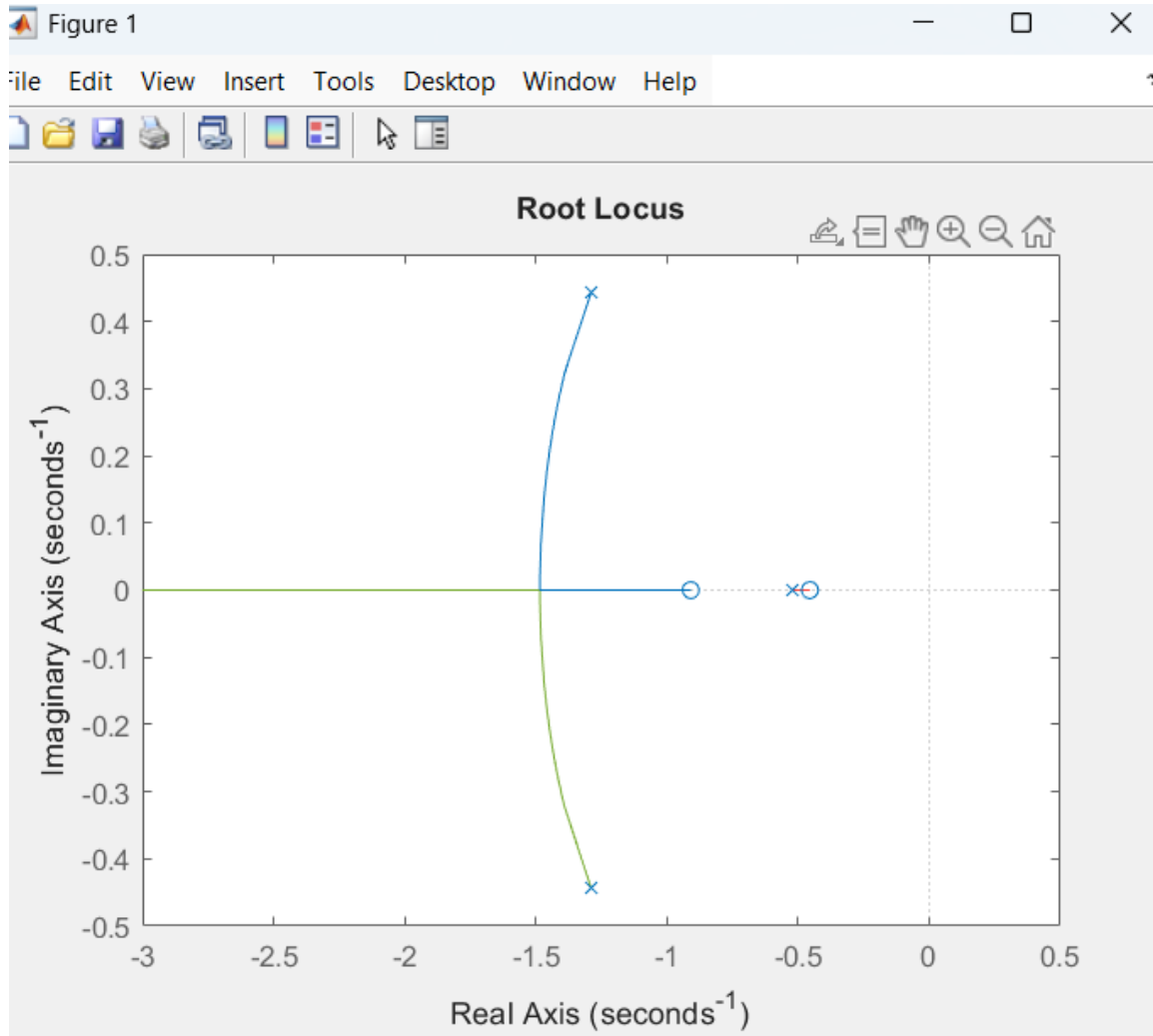


iv) Getting the following compensator:

Compensator

$$C = 1.2609 \times \frac{(1 + 2s)(1 + 2s)}{s}$$

v) After further tuning:



ompensator

$$C = 2.9275 \times \frac{(1 + 1.1s)(1 + 2.2s)}{s}$$

Conclusion

Using MATLAB, the PID controller for the DC motor was successfully designed and tuned to meet the desired performance criteria. The controller effectively improved the system's transient response, achieving faster settling time, reduced overshoot, and zero steady-state error.

Question 2:

Theory

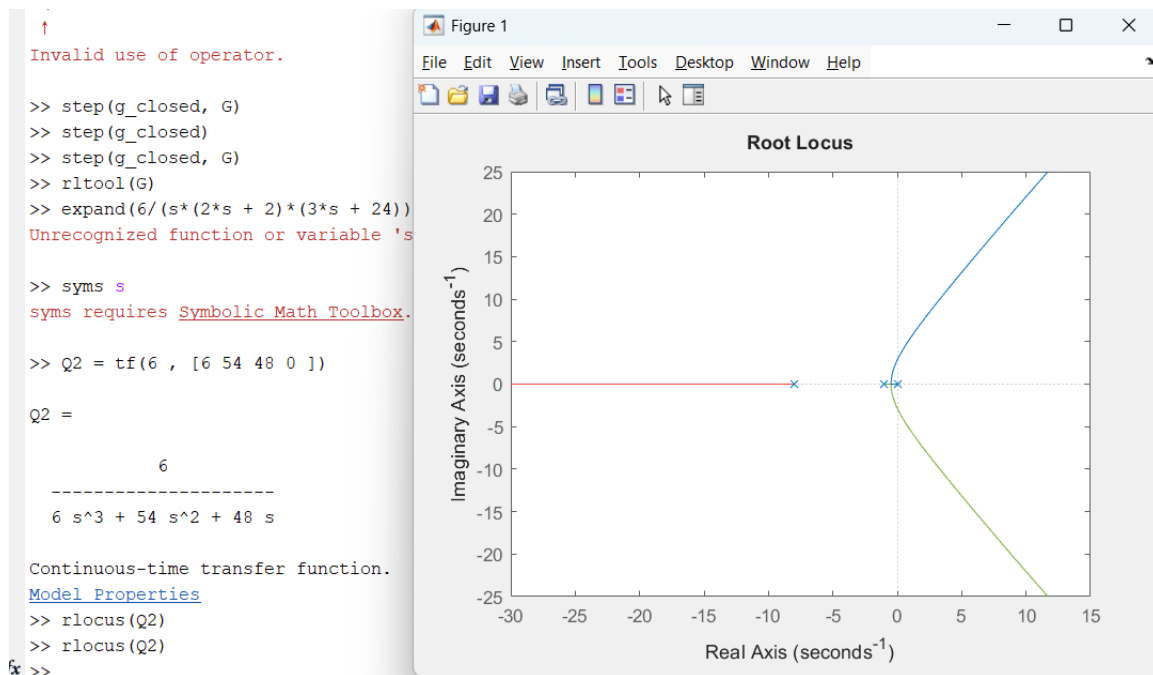
Using the requirements, $\zeta = 0.707$ and time constant = 0.5.

$\sigma = 1/\text{time constant}$, σ (Real part) = 2.

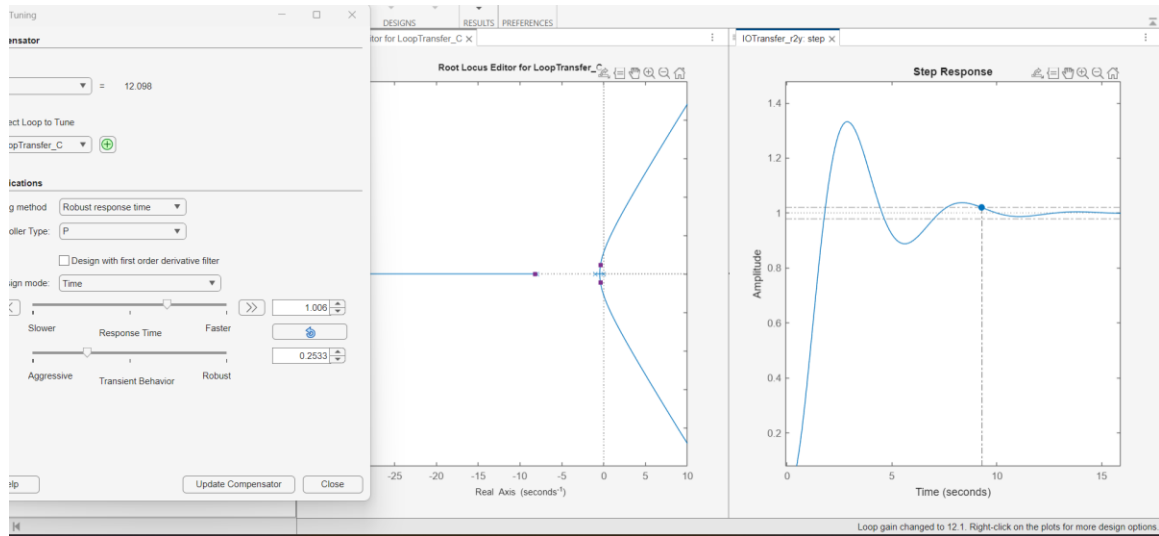
$\sigma = \zeta * \omega_n$, getting natural frequency = 2.828.

Procedures

- i) Getting the root locus for the function:

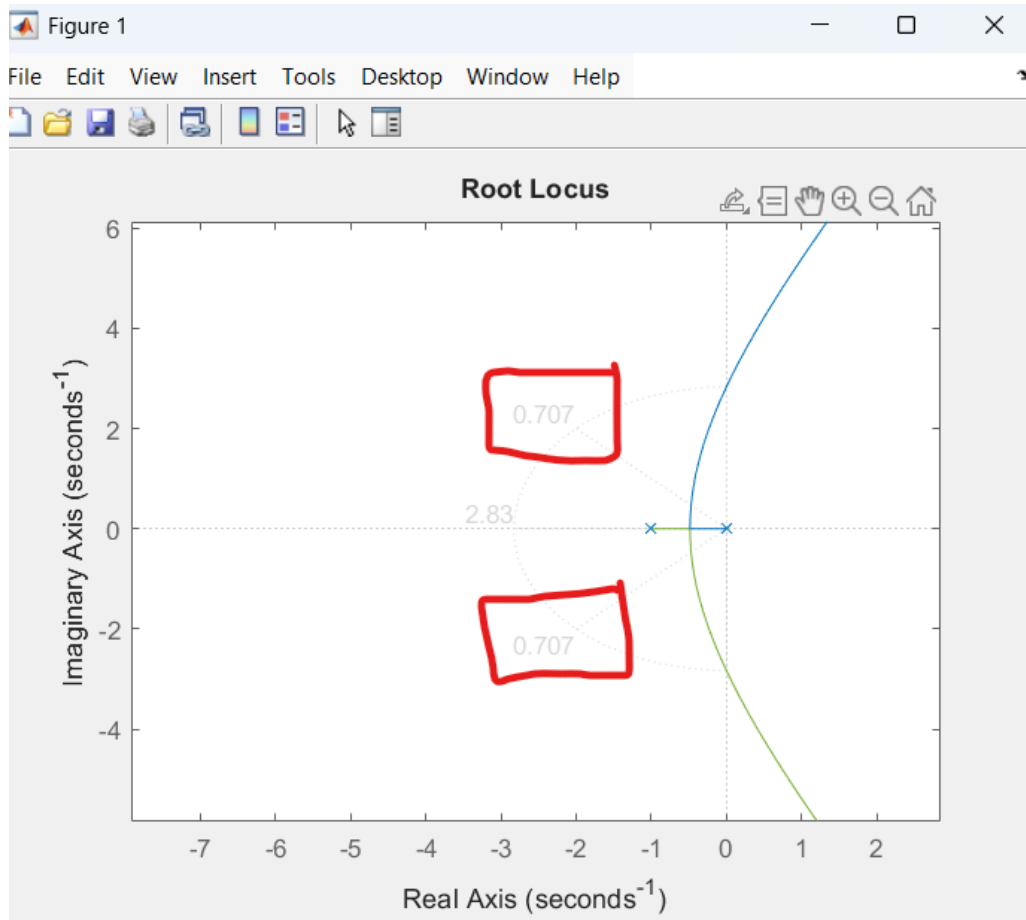


- ii) The root locus of the system is moving towards the unstable region, so we can't get 2s.

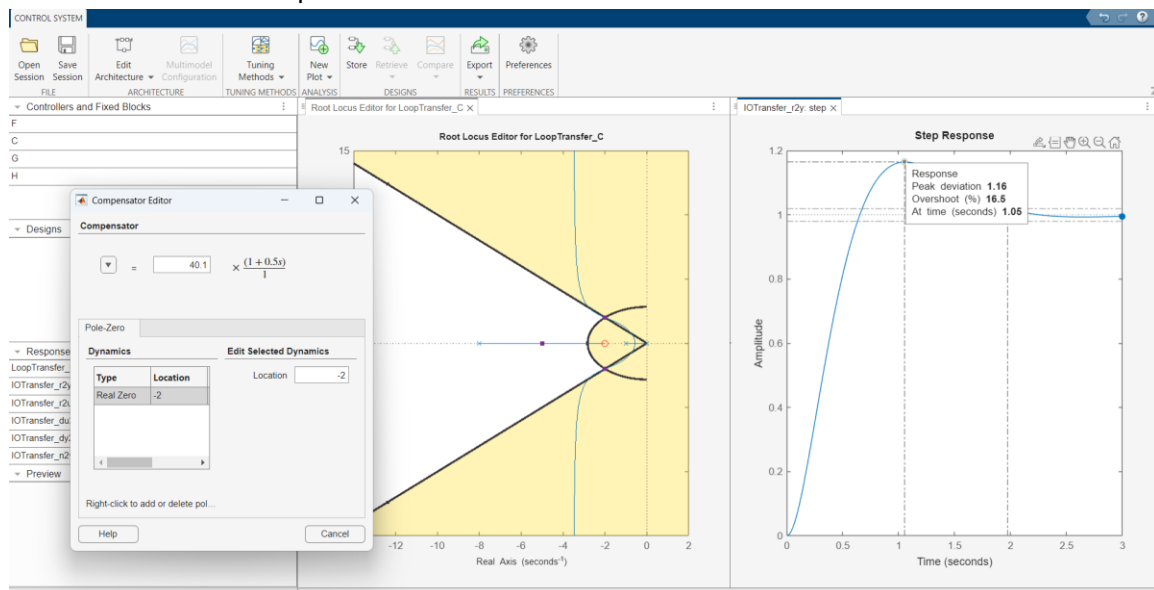


- iii) Using PD controller:

In order to the transfer function to meet its specifications, the poles should be in the marked position:



- iv) Using compensator editor to tune PD vales manually, after setting the requirements (Zeta, Omega n). the intersection is the required poles.

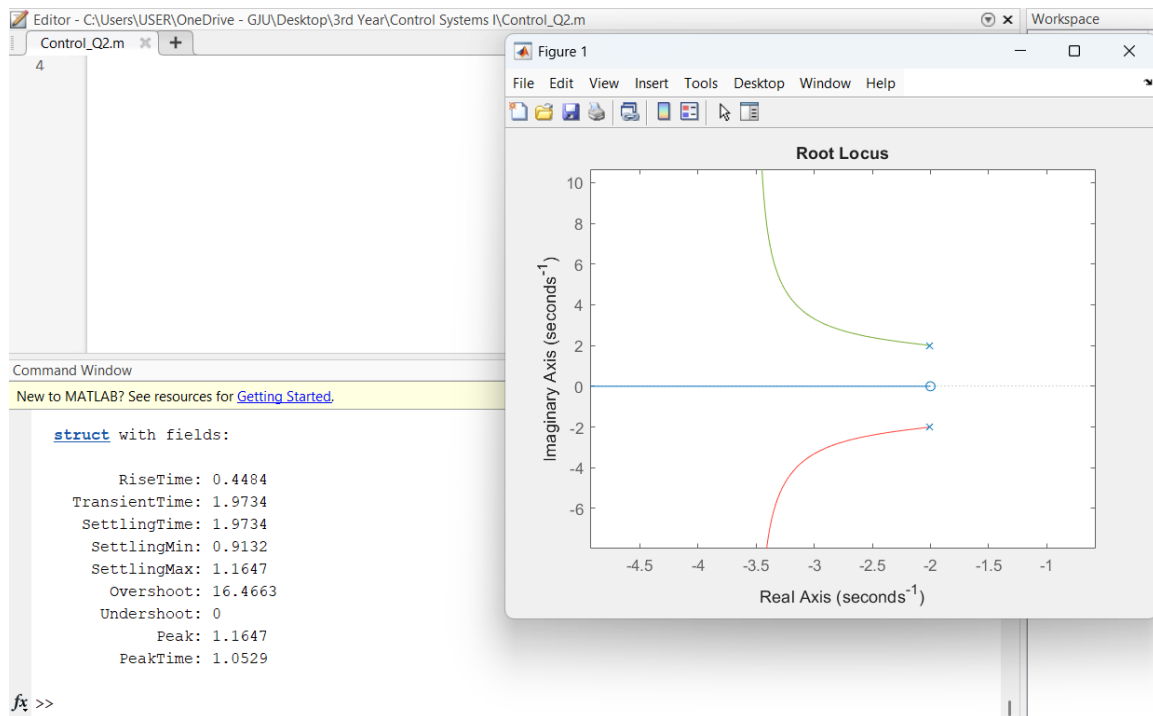


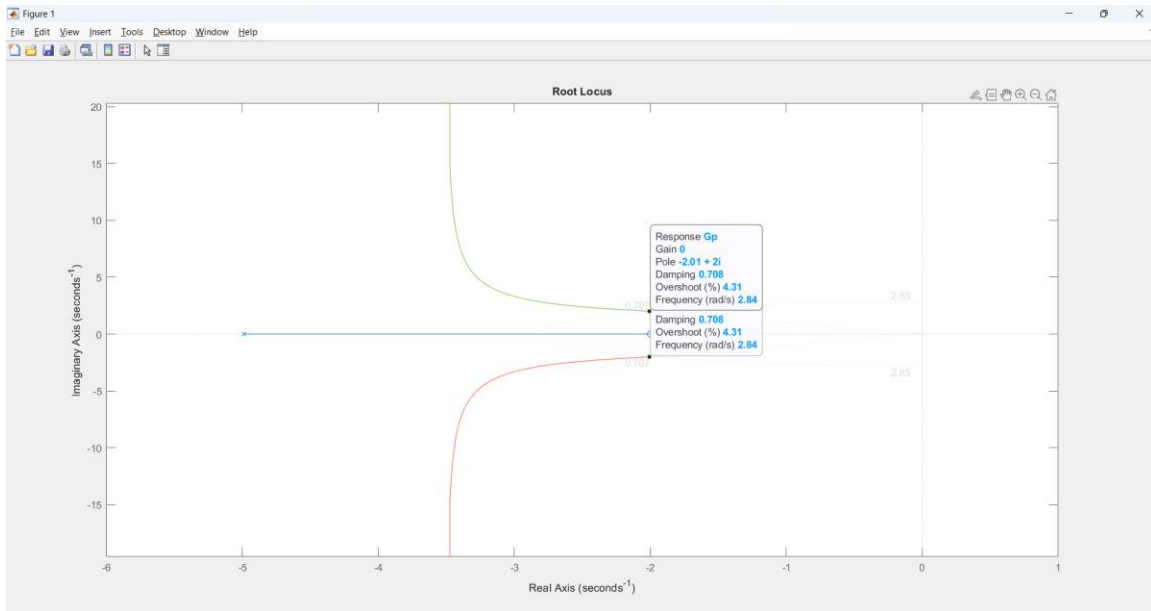
v) Setting the compensator to $K_p = 40.1$, and $K_d = 20.1$

Compensator

$$\downarrow = 40.1 \times \frac{(1 + 0.5s)}{1}$$

vi) Plotting the function with PD controller





Conclusion

Using MATLAB, proportional control alone might not achieve the desired performance, but a PD controller can effectively improve the system's response by enhancing damping and reducing the dominant time constant.